possible to test the effect of larger quantities on the organism by virtue of their radiation. A comprehensive investigation was carried out along these lines by Scott and Cook, who introduced radioactive phosphorus, produced in the Berkeley cyclotron, into the food of young hens, and then studied the haematological changes produced by the radiation. They discovered a number of interesting changes and compared them with others produced by x-rays. It is well known that blood contains two kinds of corpuscles—red and white ones. The latter are again divided into several groups, the most important of which are the polymorphonuclear leucocytes, the eosinophilic leucocytes and the basophilic leucocytes. They differ from each other in size, internal structure and staining by various substances. x-rays first make the number of lymphocytes decrease and the number of polymorphonuclear leucocytes increase, but after a short while, this multiplication of the leucocytes falls off. On the other hand, the radiation of the radioactive phosphorus introduced into the body does not affect the lymphocytes to any significant extent, but produces, instead, an appreciable permanent decrease in the number of the polymorphonuclear leucocytes. Furthermore, the corpuscles known as monocytes and the eosinophilic leucocytes are affected a little, but not strongly. The eosinophilic leucocytes and the red blood corpuscles show a slight increase in number.

This specific effect encouraged the American scientists to try to treat certain forms of leucaemia—in which the diagnosis of the disease showed considerable haematological changes—by doses of radioactive phosphorus. The phosphorus settles predominantly in the bones, and as it is well known, the red corpuscles are formed in the bone marrow, so that radioactive phosphorus is evidently capable of affecting their formation. We see how different is the effect of the radioactive phosphorus from that of x-rays, which penetrate all tissues to the same extent. The first experiments in this field are said to have been promising. In consequence of the war, however, no further information could be obtained about them. Similar experiments designed to study haematological changes have been carried out in Germany too.

Furthermore, experiments have been made with injections of radioactive lead, partly in view of the cases of lead poisoning occasionally occurring among workers, and partly with the hope of achieving therapeutic results. It was discovered that when lead is introduced into an organism, most of it is excreted very rapidly, and only a small residue remains in the liver and in the kidneys. Lead does not settle in cancerous tissue either, so that all attempts to use it for the treatment of cancer are doomed to failure. On the other hand, bismuth settles very rapidly in diseased tissue. An attempt to treat cancerous tissue by radioactive bismuth may possibly prove successful.

But all these attempts are still in their initial stages, and many years are bound to elapse before the experimental investigation emerges from the